AN X-BAND CAVITY-BACKED SLOT ANTENNA USING A VIA-HOLE ABOVE THE SLOT

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Keywords: Cavity-backed, dual resonance, slot antenna, substrate integrated waveguide(SIW), wideband

Abstract

A novel technique for bandwidth enhancement of a cavity-backed slot antenna is presented. A via-hole located above the slot creates an additional resonance at a higher frequency by shorting the effective length of the slot. The location of the via-hole can be changed to determine the second resonance frequency of the antenna. With proper placement of the via-hole, the bandwidth of cavity-backed slot antenna can be increased. A substrate integrated waveguide technology is adopted to decrease the height of the cavity, and a grounded coplanar waveguide is used as feeding network. The proposed antenna works at the X-band has a 6.8% wide bandwidth and the antenna maintains high gain during the whole band.

1 Introduction

Conventional metallic cavity backed antennas (CBAs) have been widely used for their satisfactory radiation performances[1]. The optimal distance from slot to cavity bottom is roughly one-quarter wavelength. Their bulky volumes make the conventional metallic CBAs not suitable for some practical applications. In order to achieve compact CBA, substrate integrated waveguide (SIW) technology is adopted into CBA designs. The height of the SIW cavity is far less than that of a conventional one, as according to the waveguide theory, the cavity under slot can maintain its electrical length without depending on the cavity height.

Low-profile cavity-backed slot antennas based on SIW technology have been firstly presented in [2] and [3]. A rectangle slot with a square SIW cavity was firstly presented in [2]. Its radiation is generated by the required TE120 resonance in the SIW cavity. And the dominant electric field at two sides of the slot has opposite phase, which enables the energy radiate into space when the transverse electric field across the slot. As in a conventional cavity-backed slot antenna, the resonance occurs when the slot has a half wavelength at the operating frequency. Thus the length of this rectangle slot close to a resonance slot. The antenna works at the X-band with a 1.6% bandwidth.

In [3], a non-resonance slot with an enhanced bandwidth was presented. Its length is far more than a half resonant wavelength, also the cavity sizes and length of feeding element has changed. A wider bandwidth is obtained based on two hybrid modes which are two different combinations of the TE110 and TE120 resonances. Both magnitude and phase of the fields distributing at two sides of the slot have been changed greatly. Then effective radiation can be generated from the slot. The antenna enhanced the bandwidth to 4.6% at the X-band.

The both antennas including their backed cavities and feeding elements can be completely constructed at a single substrate, which makes them easy to fabricate. While one evident drawback of these antennas is the narrow bandwidth. Thus to improve the bandwidth is very important. Similarly, impedance bandwidth of these antennas can be gradually improved by those methods such as slot with inductive or capacitive load, or adding matching circuit in feeding network.

An I-shaped slot was used for an inductive loading of the slot to decrease the resonance slot length, and a via above the slot was used for a dual resonance in [4]. The antenna works in the S-band with a 60% bandwidth.

In this letter, a new bandwidth-enhanced cavity-backed slot antenna works at the X-band is presented based on [4]. The proposed antenna has a dual resonance by using a via-hole above the slot. By adjusting the location of the via-hole, the second resonance can be moved to obtain a wider bandwidth. The antenna has 6.8% wide bandwidth.

2 Design Procedure

2.1 Antenna Configuration

Configuration of the proposed antenna is shown in Fig.1. Instead of the solid conductor cavity, SIW with via conditions ($d / d_p \geq 0.5$ and $d / \lambda_0 \leq 0.1$) is used to design the antenna at a single substrate[5]. The feeding element is similar to the geometry in [2] except for the length of it. The antenna has an I-shaped slot as an inductive loading which leads to a decrease of the slot length. The slot length is about a half of the resonance wavelength. A via-hole is used above the slot. By selecting the proper distance of the via from the center of the slot, the second resonance can be obtained without significantly influence on the first one. A wider bandwidth received as a result.

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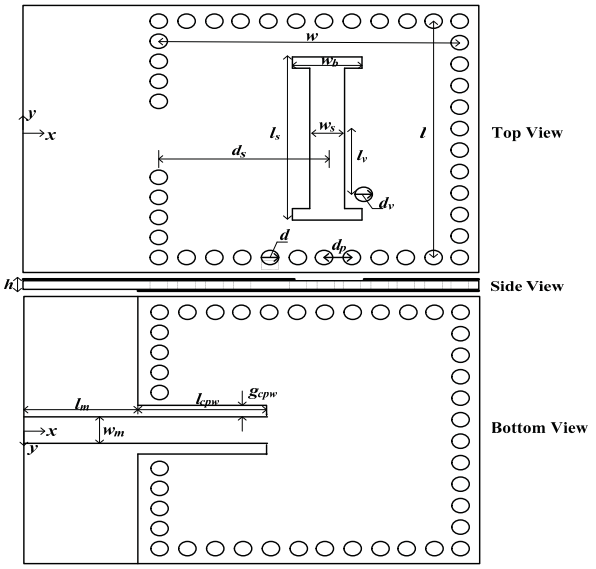


Figure 1. Geometrical configuration of the proposed antenna.

2.2 Effects of Via Location

The frequency of the first resonance is mainly determined by the cavity size and slot length. The via-hole is used above the slot to obtain a dual resonance. As shown in Fig.2, the loading effect of the via-hole is evident from the Z_{11} plot. The antenna only has one resonance (about 10GHz) when the radiation element without a via-hole. The second resonance appears (about 11.26GHz) when a via-hole is above the slot. The first resonance keeps around 10GHz with or without the via-hole when the slot length is 14.5mm which is about a half of the resonance wavelength.

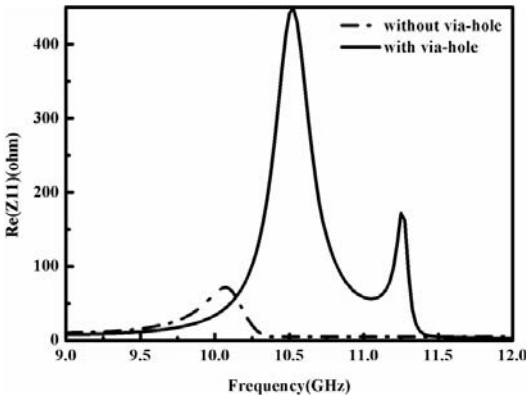


Figure 2. Real Z_{11} plot of the cavity with and without via-hole.

The second resonance, however, can be changed by moving the location of the via-hole. As shown in Fig.3, the second resonance decreases if the via-hole moves away from the center of the slot. And it does not have a significantly influence on the first resonance. By selecting the proper location for the via-hole, wideband operation can be achieved.

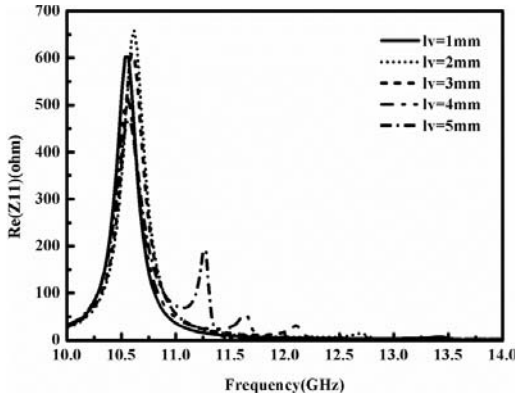


Figure 3. Variation of Real Z_{11} with change in the location of via-hole.

3 Simulation Results

For the demonstration of the proposed technique, 9.75GHz was selected as the operating frequency. Cavity size, slot length and width values were determined for the resonance at 9.75GHz. The parameters l_{cpw} , l_{ms} , m_s , g_{cpw} , which relate to the feeding element, were turned to match the antenna input impedance at 50 Ω . After determining the antenna geometry, the via-hole location and diameter are optimized for the maximum bandwidth of the antenna. The whole optimized antenna parameter values are described in Table 1. The simulation antenna was designed on the substrate with a 0.508-mm-height Rogers 5880 that had a relative permittivity of 2.2 and a loss tangent of 0.0009.

Table 1. Geometrical parameters of the proposed antenna.

parameters	units/mm	parameters	units/mm
l	17.8	w	16
d	1	d_p	1.6
l_s	16	w_s	0.5
d_v	0.4	l_v	7.6
d_s	9.5	l_{cpw}	2
l_m	2.6	g_{cpw}	1.2
w_m	2.4	h	0.508

The bandwidth of the proposed structure is very sensitive to the location of the via-hole as displayed in Fig.4. Also the first resonance changed slightly while the second resonance moves a lot when the location of the via-hole changed.

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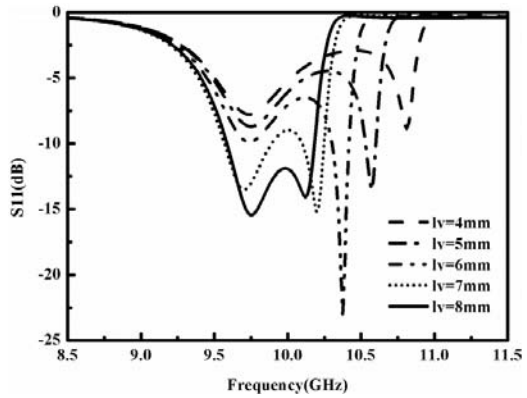


Figure 4. Variation of reflection coefficient with change in the location of via-hole.

Fig.5 shows the simulation reflection coefficients of the antenna. Regarding simulation, the -10dB bandwidth of the proposed antenna is measured to be 0.67GHz.

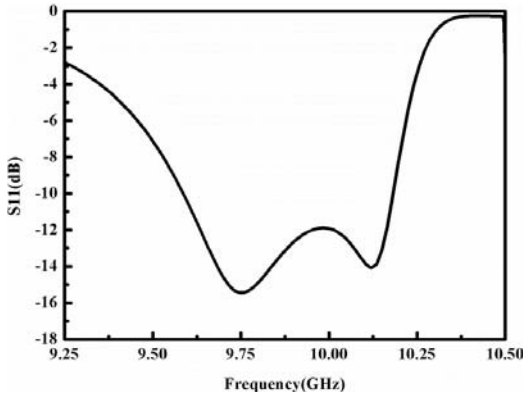


Figure 5. The simulated reflection coefficient of the proposed antenna.

The simulated antenna gain is showed in Fig.6. Although the wider-band characteristic, a flat gain higher than 6dBi is maintained over the -10 dB bandwidth.

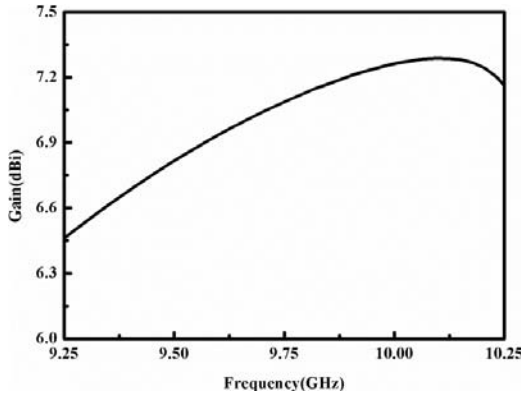
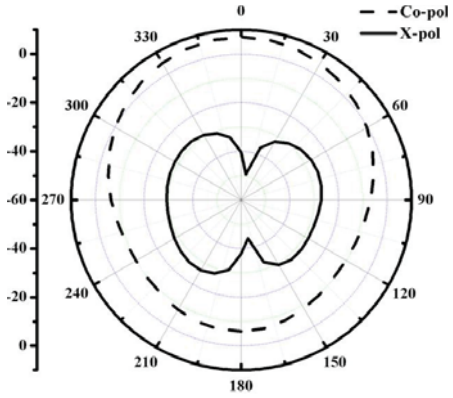
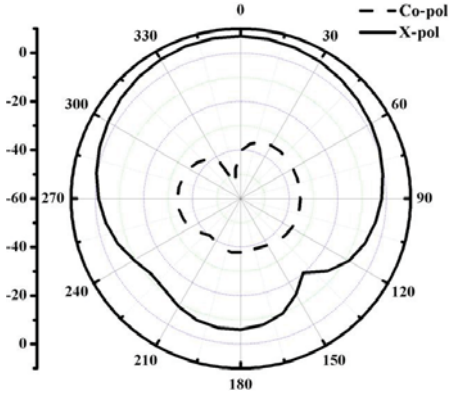


Figure 6. The simulated gain of the proposed antenna.

The simulation patterns in the E-plane(*yz* -plane) and H-plane(*xz* -plane) at 9.75GHz is shown in Fig.7. In the boresight direction, the maximum radiation was in both cut-planes. The proposed antenna also showed the very low cross-polarization levels in the two cut-planes.



(a)phi=0°



(b)phi=90°

Figure 7. The simulation patterns in the E-plane(*yz* -plane) and H-plane(*xz* -plane) at 9.75GHz.

The proposed antenna exhibits a wider-band characteristic of 6.8% which is an enhancement of the bandwidth comparing to the antennas in [1] and [2].

4 Conclusion

A technique for enhancing the bandwidth of a cavity-backed slot antenna is proposed. By shorting the effective length of the slot at the higher frequency using a single via-hole above the slot, a second resonance is achieved. By moving the second resonance frequency, the bandwidth of a cavity-backed slot antenna can be increased.

The simulation antenna has a 2.2% wider bandwidth than the antenna in [2] (4.6%). Also it maintains a high gain over the -10dB bandwidth. So it is a useful method for enhancing the bandwidth of a cavity-backed slot antenna in a limited area.

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